Project 1: Designing and Analyzing an Experiment

Econ 403B: Causal Inference & Experimentation for Decision-Making
UCLA MQE – Fall 2026 – Dr. Randall R. Rojas

Due: Friday of Week 4, 5:00PM · Weight: 20% of the course grade

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The Decision

Northline Outfitters is a direct-to-consumer outdoor gear retailer. Its product team has rebuilt the product detail page (PDP): larger imagery, a sizing guide above the fold, and customer reviews moved up from the footer. The redesign is expensive to maintain, and the VP of Product will ship it only if it raises revenue per customer by at least **$1.50 over a two-week window**. Anything smaller does not cover the engineering cost of supporting a second template.

The team ran a two-week randomized test and has handed you the log file. Your job is to tell the VP whether to ship it—and to show that the number you report can survive scrutiny.

The Data

Download `northline_pdp_test.csv` from the course website. **One row is one session.** Read the dictionary carefully before you write any code; the structure of this file matters for the analysis.

Column	Description
`user_id`	Customer identifier.
`session_id`	Session identifier.
`arm`	`control` or `treatment`.
`session_date`	Date of the session (within the two-week test window).
`revenue`	Revenue in that session, USD.
`converted`	1 if the session contained a purchase, 0 otherwise.
`device`	`mobile` or `desktop`.
`region`	Customer region.
`tenure_days`	Days since account creation, measured at assignment.
`pre_revenue_14d`	The customer's revenue in the 14 days *before* the test.
`pre_sessions_14d`	The customer's session count in the 14 days *before* the test.

What to Do

Part A. Design (do this before you look at the outcome)

A1. State the estimand. Define the treatment, the unit, the outcome, and the target population. Write the potential outcomes $Y_i(1)$ and $Y_i(0)$ for your chosen unit in one line of notation, and give the estimand as an expectation. Say which estimand the VP's decision actually requires—ATE, ATT, or something else—and why. State SUTVA in the language of this problem and name one thing that would violate it here.

A2. Draw the assignment. Give a small DAG (five nodes is plenty) containing the treatment, the outcome, at least one pre-treatment customer characteristic, and one *post*-treatment variable such as time on page. Use it to explain in two or three sentences why random assignment identifies the effect, and name one variable in the dataset that you must **not** condition on. Justify that exclusion with the graph, not with intuition.

A3. Choose the randomization unit. The team could have randomized by session, by user, or by region. State which unit was in fact used—verify this in the data rather than assuming it—and explain what the alternatives would have cost in terms of bias and variance. One paragraph.

A4. Power and MDE. Using only `pre_revenue_14d` as a stand-in for the outcome's dispersion, compute:

(a) the minimum detectable effect at the realized sample size, with $\alpha = 0.05$ and power 0.80;

(b) the sample size that would have been required to detect the VP's \$1.50 threshold at the same α and power.

Then answer the question that matters: *was this test capable of answering the question it was run to answer?* If the MDE exceeds \$1.50, say so plainly and carry that fact into your recommendation.

Part B. Analysis

B1. Validity checks. Before estimating anything, run and report:

- a sample ratio mismatch test (is the split what the team intended?);
- balance on `pre_revenue_14d`, `pre_sessions_14d`, `tenure_days`, `device`, and `region`—report standardized differences, not just p-values;
- a check for missingness or duplicated identifiers.

Two sentences on what these checks can and cannot tell you. A balance table that passes is not evidence that the experiment was run correctly.

B2. Naive estimate. Estimate the treatment effect the way an analyst in a hurry would: a difference in means on the file as delivered, with the usual heteroskedasticity-robust standard error. Report the point estimate, standard error, and 95% confidence interval. *Do not skip this step*—you will need it for B3.

B3. Diagnose the failure mode. There is a specific problem with the naive estimate in B2. Find it, demonstrate it with evidence from the data rather than asserting it, explain in plain language why it biases the inference and in which direction, and produce a corrected estimate. Report the naive and corrected standard errors side by side and state how much the naive one understates the true uncertainty. This part is worth the most points in the project; a correct number with no diagnosis earns nothing.

B4. Variance reduction. Using your corrected specification, re-estimate the effect with CUPED (or equivalent covariate adjustment) using the pre-period outcome. Report the estimate, the percentage reduction in the variance of the estimator, and the MDE that this test *would* have had if CUPED had been planned from the start. Compare that to your answer in A4.

B5. Randomization inference. Compute a permutation-based p-value for the sharp null of no effect for any unit, permuting at the correct level. Compare it to your analytical p-value and comment on any discrepancy in one or two sentences.

Part C. The Recommendation

C1. Write **one paragraph, at most ∼200 words**, addressed to the VP of Product, who has no statistical training. It must state: the recommendation (ship, do not ship, or keep testing), the estimand it rests on, the single assumption that would most threaten it, and the one thing that would change your mind. No equations, no jargon, no hedging that leaves the VP without an answer.

C2. In two or three sentences: name the threat you would worry about most if this redesign were rolled out to 100% of traffic rather than 50%, and say what you would measure to detect it.

Deliverables and Format

- One PDF, produced from Quarto or a Jupyter notebook, **5-6 pages maximum** excluding code appendix and figures. Prose over output dumps; a wall of unannotated regression tables is not an answer.
- The code that produced every number, either inline in the source document or as an accompanying `.ipynb` / `.qmd`. **Submissions without code receive no credit**, per the syllabus.
- Set a seed. Your results must reproduce.
- Course policy applies: every empirical answer states the estimand, the identifying assumption(s) it relies on, and at least one diagnostic or robustness check. Numerical results reported without this reasoning receive no credit, even when the point estimate is correct.
- Late work is penalized 1 point per minute.

Scope note. Everything required here is covered by the end of Week 2. Nothing in this project requires panel methods, matching, or machine learning. Budget roughly six to eight hours. If you are spending substantially more than that, you have probably wandered into a method the project does not ask for.

Grading

Component		Points
A1–A2	Estimand, potential outcomes, SUTVA, DAG and bad-control argument	20
A3–A4	Randomization unit; power and MDE, correctly interpreted	15
B1	Validity checks and what they establish	10
B2–B3	Naive estimate, diagnosis of the failure mode, corrected estimate	30
B4–B5	CUPED and randomization inference	15
C1–C2	Recommendation to a non-technical decision maker	10
Total		**100**

Points in B2–B3 are awarded for the diagnosis and the evidence behind it, not for arriving at the corrected number. A submission that reports the right estimate without identifying why the naive one was wrong will score near zero on that component.